

Fitting Covariance vs Response

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We fit connectivity in two ways: by fitting the raw responses, and fitting the covariance of the responses. How are the resulting connectivities related?

When we fit the responses, our loss is

$$\mathcal{L}_{\text{resp}}(\mathbf{Z}) = \frac{1}{2} \|\mathbf{Y} - \mathbf{Z}\mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{Z} - \mathbf{I}\|_F^2,$$

while when fitting covariances, we minimize

$$\mathcal{L}_{\text{cov}}(\mathbf{Z}) = \frac{1}{2} \|\mathbf{Y}^T \mathbf{J} \mathbf{Y} - \mathbf{X}^T \mathbf{Z}^T \mathbf{J} \mathbf{Z} \mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{Z} - \mathbf{I}\|_F^2.$$

To relate these, we use the decomposition into mean-subtracted and mean components.

$$\mathbf{X} = \mathbf{J}\mathbf{X} + (\mathbf{I} - \mathbf{J})\mathbf{X} \triangleq \tilde{\mathbf{X}} + \bar{\mathbf{X}}.$$

Notice that these two components are orthogonal,

$$\langle \tilde{\mathbf{X}}, \bar{\mathbf{X}} \rangle = \text{tr}(\tilde{\mathbf{X}}^T \bar{\mathbf{X}}) = \text{tr}(\mathbf{X}^T \mathbf{J}^T (\mathbf{I} - \mathbf{J}) \mathbf{X}) = \text{tr}(\mathbf{X}^T \mathbf{0} \mathbf{X}) = 0.$$

That means that we can decompose our response loss into two independent components,

$$\begin{aligned} \mathcal{L}_{\text{resp}}(\mathbf{Z}) = \mathcal{L}_{\text{resp}}(\mathbf{J}\mathbf{Z}, \bar{\mathbf{Z}}) &= \frac{1}{2} \|\mathbf{J}\mathbf{Y} - \mathbf{J}\mathbf{Z}\mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{J}\mathbf{Z} - \mathbf{J}\|_F^2 \\ &\quad + \frac{1}{2} \|\bar{\mathbf{Y}} - \bar{\mathbf{Z}}\mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\bar{\mathbf{Z}} - \bar{\mathbf{I}}\|_F^2. \end{aligned}$$

We can do the same thing for the covariance loss,

$$\begin{aligned} \mathcal{L}_{\text{cov}}(\mathbf{Z}) = \mathcal{L}_{\text{cov}}(\mathbf{J}\mathbf{Z}, \bar{\mathbf{Z}}) &= \frac{1}{2} \|(\mathbf{J}\mathbf{Y})^T \mathbf{J} \mathbf{Y} - \mathbf{X}^T (\mathbf{J}\mathbf{Z})^T (\mathbf{J}\mathbf{Z}) \mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{J}\mathbf{Z} - \mathbf{J}\|_F^2 \\ &\quad + \frac{\lambda}{2} \|\bar{\mathbf{Z}} - \bar{\mathbf{I}}\|_F^2. \end{aligned}$$

From this, we immediately see that the regularization in the covariance case will send $\bar{\mathbf{Z}}$ to $\bar{\mathbf{I}}$.

Therefore, the aspect of the connectivity to compare between the two losses is $\mathbf{J}\mathbf{Z}$. We can polar decompose this into the product of a pure rotation $\mathbf{R}$ and a positive-semi-definite stretch $\mathbf{S}$,

$$\mathbf{J}\mathbf{Z} = \mathbf{R}\mathbf{S}.$$

Since $(\mathbf{J}\mathbf{Z})^T (\mathbf{J}\mathbf{Z}) = \mathbf{S}^T \mathbf{R}^T \mathbf{R} \mathbf{S} = \mathbf{S}^T \mathbf{S}$, we can see that the covariance fit only determines the stretch component, while the response fit determines both.

Therefore, when comparing connectivity solutions, we should use $\mathbf{R}$ and $\mathbf{S}$ computed from $\mathbf{J}\mathbf{Z}$, not $\mathbf{Z}$, when determine the effects of each component.

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